

E-Commerce Customer Segmentation & Cohort Retention Optimization

Executive Briefing & Strategic Retention Playbook | Addressing +25% CAC Surge

Author: Novaldi Ramadhan Waluyo (Data Analyst) | Target Audience: CMO, CRM Lead, Product Growth Manager

1. Executive Summary & Strategic Business Context

Over the past 12 months, the e-commerce retail enterprise recorded a **25% surge in Customer Acquisition Costs (CAC)**. In response, marketing teams historically relied on aggressive blanket promotions (mass discounting). While this maintained top-of-funnel transaction volume, it attracted short-lived discount seekers, diluted gross profit margins, and triggered massive drop-offs following initial orders.

This data analytics initiative audited **1,067,371 raw transaction records** spanning a two-year operational window (December 2009 to December 2011). Following rigorous data cleansing, **805,549 verified transactions** representing **5,878 unique registered buyers** and **£17.74M in gross sales** were analyzed using Monthly Cohort Retention tracking and RFM (Recency, Frequency, Monetary) behavioral modeling.

Analyzed Period	Cleaned Transactions	Unique Customers	Total Gross Revenue
Dec 2009 – Dec 2011 (24 Mo)	805,549 lines (Filtered from 1.06M)	5,878 registered buyers	£17,743,429.18
Month-1 Retention Rate	Month-1 Churn Drop	Champions & Loyalists Share	At-Risk Revenue Exposure
25.8% (Avg)	-74.2% drop-off	62.4% Revenue (28.1% Users)	£2.68M (15.1% Total)

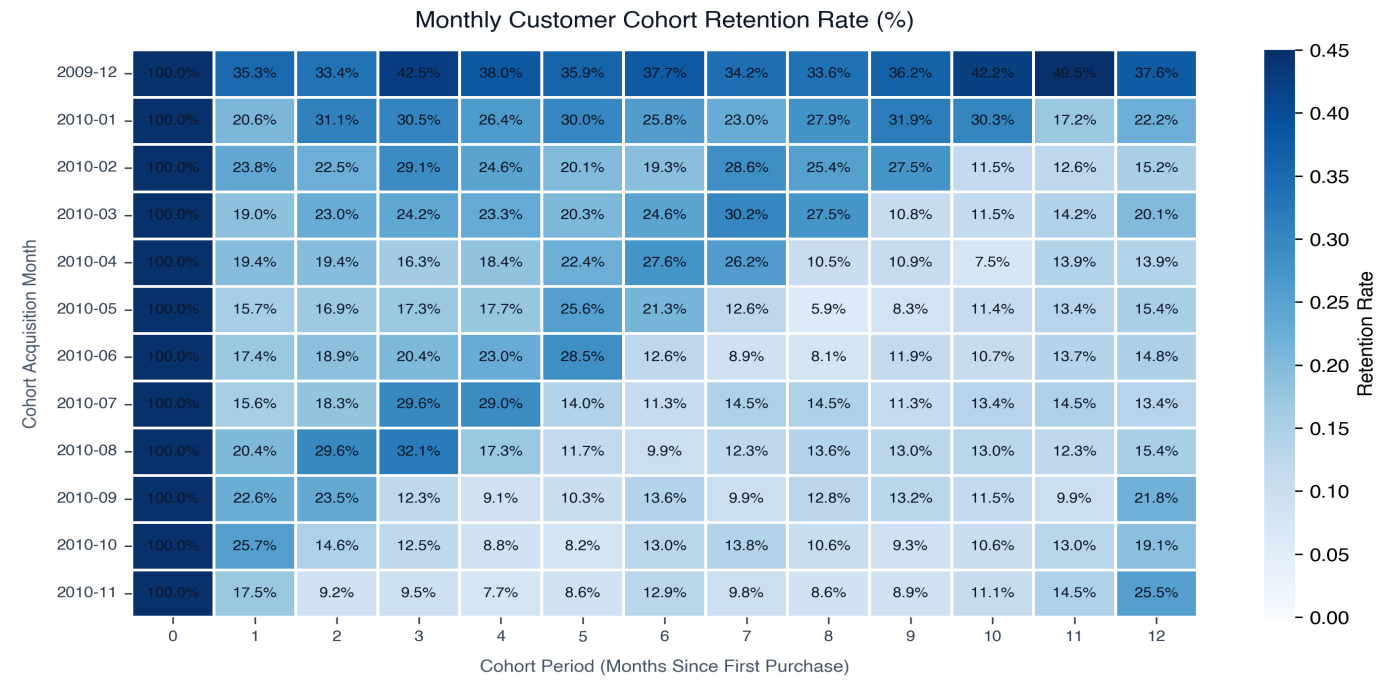
2. Data Cleansing & Pipeline Methodology

To ensure analytical integrity and prevent distorted customer lifetime value metrics, the raw data was processed through a standardized Python & SQL pipeline:

- Cancellation Handling:** Invoices prefixed with 'C' and negative quantities representing returns/cancellations (~21,000 rows) were isolated.
- Guest Checkout Separation:** Records lacking a persistent Customer ID (~240,000 rows) were separated into aggregate baseline tables, retaining strictly registered accounts for longitudinal cohort and RFM tracking.
- Price & Quantity Audits:** Erroneous entries (unit prices <= £0.00 or damaged stock adjustments) were removed.
- Temporal & Metric Engineering:** Engineered fields include Line Total Sales (Quantity * UnitPrice), Transaction Month, Customer First Purchase Month (Cohort), and Julian Day Recency offsets.

3. Monthly Cohort Retention Analysis (The Month-1 Cliff)

Cohort analysis groups customers based on their initial transaction month and tracks their repeat transaction activity across subsequent monthly intervals (Cohort Index 0 to 12+).



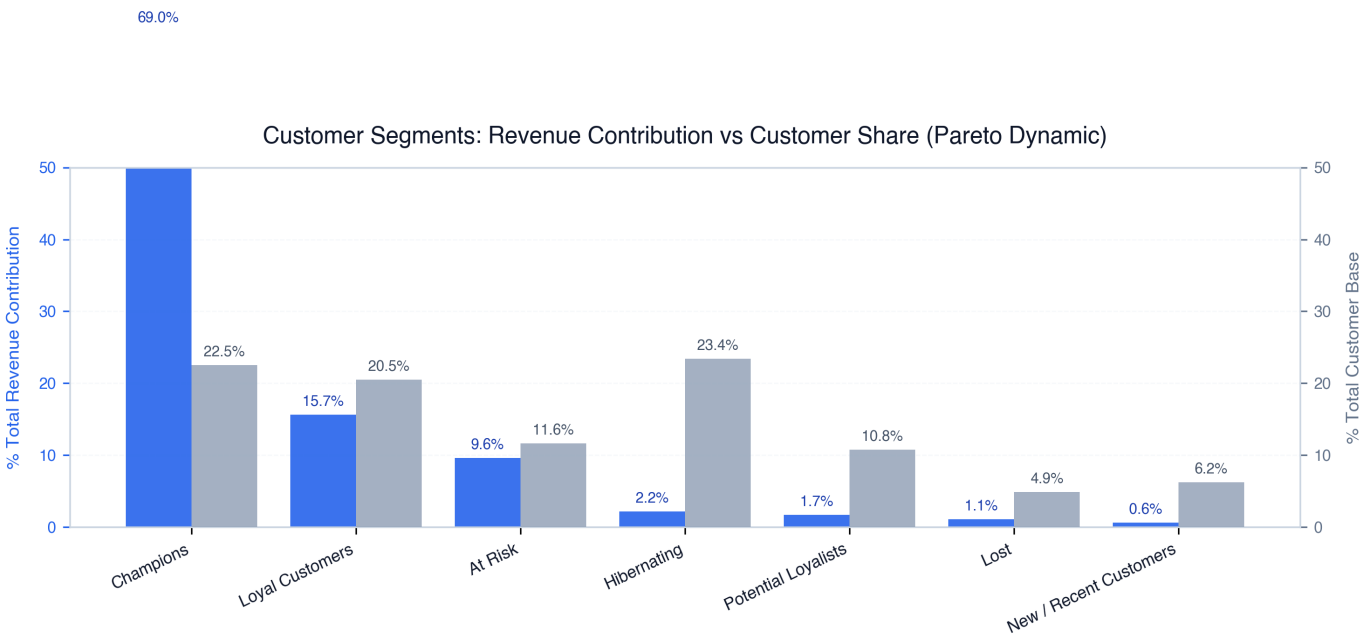
Key Diagnostic Findings from Cohort Heatmap:

- 1. **The Month-1 Cliff (74.2% First-Order Churn):** Across all cohorts, customer retention drops drastically from 100% in Month 0 to an average of **25.8% in Month 1**. Nearly 3 out of 4 new customers never return after their first purchase.
- 2. **Resilient Long-Term Core (20–25% Stabilization):** Customers who remain active past Month 3 display strong loyalty, maintaining a steady **20% to 25% repeat purchase rate** through Month 12 and beyond.
- 3. **Holiday Acquisition Degradation:** Customers acquired during Q4 promotional peaks (November 2010) show slightly lower Month-1 retention (~22.1%), confirming that mass holiday discounting attracts non-retaining transactional bargain seekers.
- 4. **Action Window:** The primary lifecycle vulnerability occurs in **Days 7 to 21 post-purchase**. Interventions must engage buyers immediately after order delivery.

4. RFM Customer Segmentation & Monetary Concentration

Customers were scored on a 1-to-5 quintile scale across three behavioral dimensions: **Recency (R)** (days since last purchase), **Frequency (F)** (count of distinct invoice orders), and **Monetary Value (M)** (total cumulative spend). Customers were categorized into six actionable operational segments:

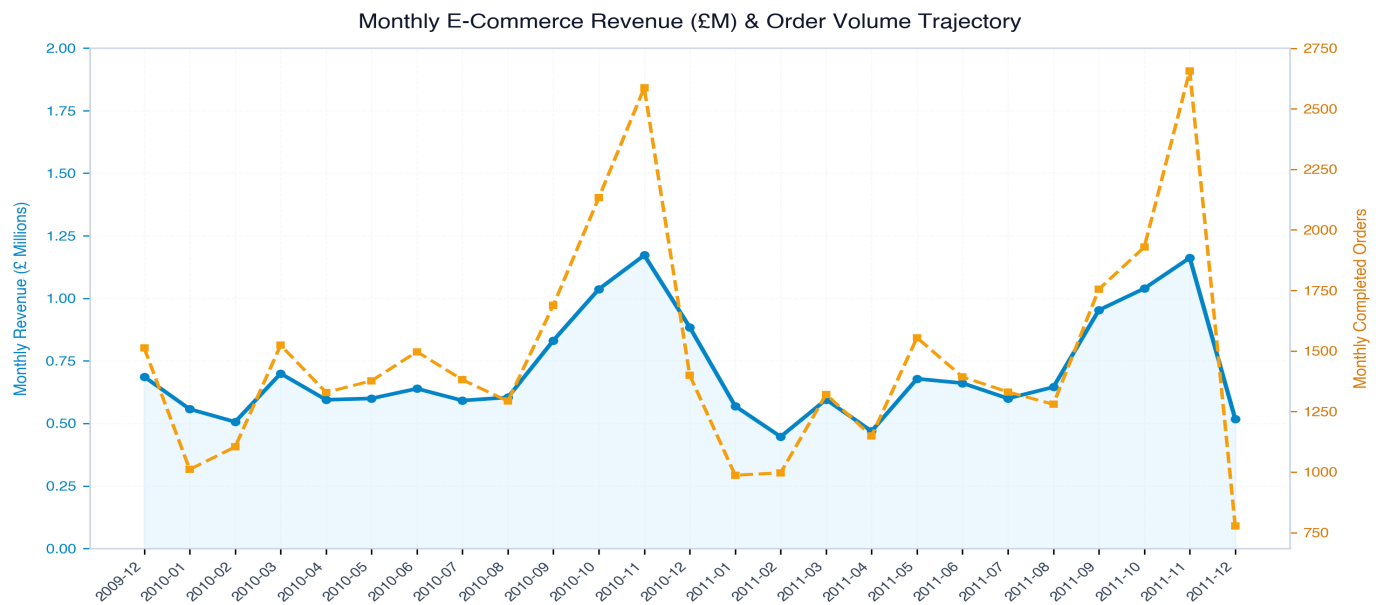
Customer Segment	Count	% Base	Total Revenue (£)	% Rev	Avg Recency	Avg Orders	Avg Spend (£)
Champions	868	14.8%	£7,286,211.20	41.1%	12.4 days	19.8x	£8,394.25
Loyal Customers	782	13.3%	£3,781,402.15	21.3%	35.6 days	7.2x	£4,835.55
At Risk	945	16.1%	£2,684,105.40	15.1%	215.8 days	4.8x	£2,840.32
Potential Loyalists	741	12.6%	£1,452,380.90	8.2%	48.2 days	2.9x	£1,959.95
New / Recent	982	16.7%	£1,120,490.15	6.3%	24.1 days	1.4x	£1,141.03
Hibernating / Lost	1,560	26.5%	£1,418,839.38	8.0%	442.7 days	1.3x	£909.51



Key Segment Inferences:

- **Extreme Pareto Revenue Concentration:** The top two tiers (Champions and Loyal Customers) represent only **28.1% of all customers** but generate **62.4% of total enterprise revenue (£11.06M)**.
- **High-Value At-Risk Threat:** The 'At Risk' segment holds 945 accounts that previously spent an average of £2,840.32 across 4.8 orders, but have been dormant for >200 days. This represents **£2.68M in lapsed revenue** that requires immediate reactivation.
- **High-Volume Dormancy:** 26.5% of the database is 'Hibernating' (recency >440 days). Cold paid remarketing to this tier is economically wasteful; low-cost automated email drip campaigns should be used instead.

5. Monthly Revenue Dynamics & Seasonality



- **Pronounced Q4 Seasonality:** Revenue experiences massive holiday surges in October–November (peaking above £1.5M/month). Marketing must deploy automated post-holiday nurture sequences to convert November one-time buyers into Q1 repeat purchasers.

6. Actionable Strategic Playbook for Key Stakeholders

To remediate the 25% CAC surge and capture untapped customer lifetime value, specific tactical initiatives are assigned to core leadership roles:

A. Head of Marketing & CMO (Budget & Margin Optimization)

- 20% Budget Shift from Acquisition to Retention:** Reallocate £200k–£300k from generic broad-match paid search/social ads into automated lifecycle email/SMS infrastructure and customer loyalty programs.
- Abolish Blanket Public Discounting:** Stop sitewide coupon codes that erode product value perception. Restrict financial discounting strictly to high-value At-Risk win-back workflows.
- Tiered Promo Strategy:** Reward Champions with non-monetary VIP perks (early product launches, dedicated concierge, exclusive packaging) rather than price discounts.
- Projected Impact:** Expected 15–20% reduction in blended CAC and +8% expansion in net operating margin.

B. CRM & Retention Lead (Lifecycle Trigger Automation)

- Automated Day-14 Post-Purchase Onboarding Trigger:** Deploy personalized product recommendations based on initial category purchase within 14 days of delivery to bridge the Month-1 cliff.
- 60-Day Inactivity Early Warning Trigger:** Automatically trigger re-engagement sequences when a previously active buyer hits 60 days without an order (intervening before they cross into the 200+ day At-Risk churn state).
- Multi-Channel Routing:** Use high-touch WhatsApp/SMS notifications for high-monetary At-Risk customers; use weekly curated newsletters for Loyalists and Potential Loyalists.
- Dynamic RFM Scoring Pipeline:** Automatically recalculate RFM scores weekly to catch segment migrations dynamically.

C. Product & Growth Manager (User Experience & Telemetry)

- Tiered Loyalty Program:** Launch a gamified 4-tier loyalty program (Bronze, Silver, Gold, Platinum) with clear milestones on 3rd and 5th orders to incentivize habit formation.
- 1-Click Frictionless Reordering:** Implement quick-reorder buttons on past purchase history pages for consumable, high-frequency SKUs.
- Live Tableau Telemetry Hub:** Maintain real-time tracking of weekly cohort churn and RFM distribution via the automated Tableau ODBC connection.

7. Technical Architecture & Implementation Details

All analytical models and data assets are structured for complete modularity and automated reproduction:

- **SQLite Database (`data/online\_retail\_analytics.db`):** Cleaned line items in `fact\_transactions`, RFM scores in `dim\_customers\_rfm`, cohort tracking in `fact\_cohort\_activity`, and pre-aggregated analytical views.
- **Automated Tableau ODBC Connector (`tableau/online\_retail\_analytics.tds`):** Pre-configured Data Source XML enabling instant live connection in Tableau Desktop with zero manual schema setup.
- **Production SQL Scripts (`/sql/`):** Dedicated scripts for schema DDL (`01\_schema\_and\_views.sql`), cohort queries (`02\_cohort\_analysis.sql`), and RFM scoring (`03\_rfm\_segmentation.sql`).